Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

Design and Performance Analysis of a CMOS Inverter using MAGIC & SPICE

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Final Project Report

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Exercise 1

The physical layout of the CMOS inverter, designed using the **MAGIC VLSI** tool, is illustrated in Figure 1. The design strictly adheres to minimum area constraints, featuring optimized transistor sizing and precise layer alignment to ensure geometric integrity. Specifically, the NMOS and PMOS transistors were sized with $W = 3\mu m$ and $L = 2\mu m$ as per the assignment specifications.

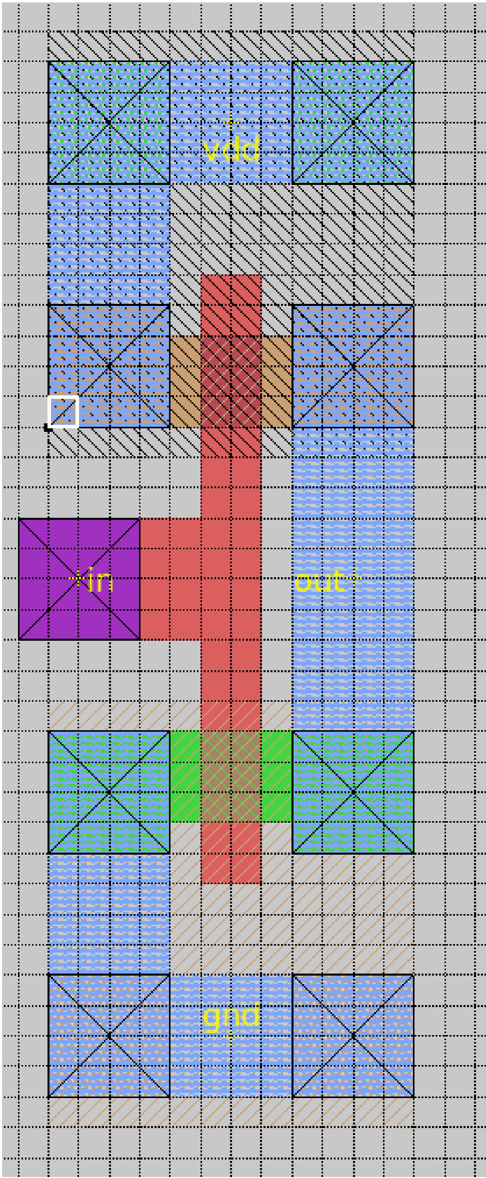

Figure 1: Physical layout implementation of the CMOS inverter in Magic VLSI.

To bridge the gap between physical design and circuit verification, a parasitic extraction process was performed. By executing the commands `extract all` followed by `ext2spice` within the MAGIC environment, the layout was converted into a comprehensive **SPICE netlist**. This generated file, saved as $askisi\_1.spice$, contains the structural connectivity, device parameters, and transistor models (CMOSN, CMOSP) required for subsequent electrical simulation and timing analysis.

The generated SPICE netlist, as extracted from the physical layout, is presented below:

```
.option scale=1u

M1000 out in gnd gnd nfet w=3 l=2
+ ad=19 pd=18 as=19 ps=18

M1001 out in vdd vdd pfet w=3 l=2
+ ad=19 pd=18 as=19 ps=18
```

A detailed structural analysis of the netlist confirms that all transistor terminals (Drain, Gate, Source, and Body) have been correctly mapped according to the CMOS inverter schematic. Specifically:

- NMOS Transistor (M1000): The drain is routed to the `out` node, the gate to the `in` node, and both the source and body (bulk) are tied to `gnd`.
- PMOS Transistor (M1001): Symmetrically, the drain is connected to the `out` node and the gate to the `in` node, while the source and body are tied to the supply voltage `vdd`.

The extraction results verify that the layout maintains full electrical connectivity with the intended logic gate and that the physical dimensions ($W = 3\mu m$, $L = 2\mu m$) were accurately translated into the SPICE environment.

Exercise 2

The second phase of the assignment involves the functional verification of the inverter through SPICE simulation and the characterization of its dynamic performance. According to the design specifications, the input signal is defined with rise and fall times of $t_{rise} = t_{fall} = 200\,ps$.

To accurately model the input waveform in the SPICE environment, the signal was constructed such that the transient duration between the 10% and 90% thresholds of the supply voltage ($V_{DD} = 2.5V$) corresponds to the required $200\,ps$.Specifically, the time interval $\Delta t = x_2 - x_1$ was calculated based on the following voltage ordinates:

- $y_1 = 10\% \cdot 2.5V = 0.25V$
- $y_2 = 90\% \cdot 2.5V = 2.25V$

The simulation utilizes the `0.25-models` for the CMOSN and CMOSP transistors, with a transient analysis step set to $10\,ps$ to ensure high-resolution capture of the switching characteristics.

To define the Piecewise Linear (PWL) voltage source in SPICE, a linear interpolation approach was employed to determine the exact coordinates of the transition. Assuming the transition points for 10% and 90% of V_{DD} are located at $x_1 = 200\,ps$ and $x_2 = 400\,ps$ respectively (yielding the required $\Delta t = 200\,ps$), we define the slope of the rising edge.

Given that the signal transition is linear from 0% to 100% of the supply voltage, the slope (λ_1) is calculated as follows:

$$\lambda_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2.25V - 0.25V}{400ps - 200ps} = \frac{2.0V}{200ps} = 0.01\,V/ps \tag{1}$$

Using the point-slope form, the equation for the rising edge is derived:

$$y - 2.25 = \lambda_1(x - 400) \implies y = 0.01x - 1.75 \tag{2}$$

By solving this linear equation, we can precisely determine the time coordinates for the $0V$ and $2.5V$ points required for the SPICE `PWL` card, ensuring that the simulated rise time strictly adheres to the $200\,ps$ specification.

Following the rising transition, the input signal reaches its peak value of $V_{in} = 100\% \cdot V_{DD} = 2.5V$. To ensure that the circuit reaches a stable logic state and to allow sufficient time for the output to respond, the input voltage is maintained at this level for a specific duration.

The steady-state phase is defined between the time points $t_{pulse\_start} = 425\,ps$ and $t_{pulse\_end} = 1000\,ps$. This fixed-voltage interval is crucial for the subsequent characterization of the propagation delay and the output settling time, as it prevents any overlap between the rising and falling edge transients during the simulation analysis.

Adopting a consistent methodology for the trailing edge of the input pulse, we derive the linear equation for the falling transition. Given that the fall time duration is specified to be identical to the rise time ($\Delta t = 200\,ps$), the absolute value of the slope remains constant. However, due to the descending monotonicity of the signal, the slope coefficient takes a negative value.

The slope (λ_2) for the falling edge is calculated as follows:

$$\lambda_2 = \frac{y_{final} - y_{initial}}{x_2 - x_1} = \frac{0.25V - 2.25V}{200\,ps} = -0.01\,V/ps \tag{3}$$

By applying this slope to the timing constraints of the pulse, the falling edge is fully defined, ensuring symmetry between the rising and falling transients of the input waveform.

Using again the point-slope form, the equation for the falling edge is derived:

$$y - 2.5 = \lambda_1(x - 1000) \implies y = -0.01x + 12.5 \tag{4}$$

By integrating the linear equations derived for the rising and falling edges, we constructed the final input waveform (V_{in}) used to drive the CMOS inverter. The synthesized signal is characterized by its symmetric transition periods and stable steady-state intervals, ensuring a reliable benchmark for timing analysis.

The primary characteristics of the input signal are summarized in Table 1, while Figure 2 provides a detailed graphical representation of its temporal behavior.

Table 1: SPICE PWL Input Signal Coordinates

Point	**Time (t)**	**Voltage (V)**	**Description**
P_1	$175\,ps$	$0\,V$	Start of Rising Edge
P_2	$425\,ps$	$2.5\,V$	Stabilization at Logic '1' (V_{DD})
P_3	$1000\,ps$	$2.5\,V$	Initiation of Falling Edge
P_4	$1250\,ps$	$0\,V$	Return to Logic '0' (GND)

Based on these coordinates, the input signal provides a clear trigger for the inverter, allowing for the accurate measurement of propagation delays from the 50% threshold of the transition.

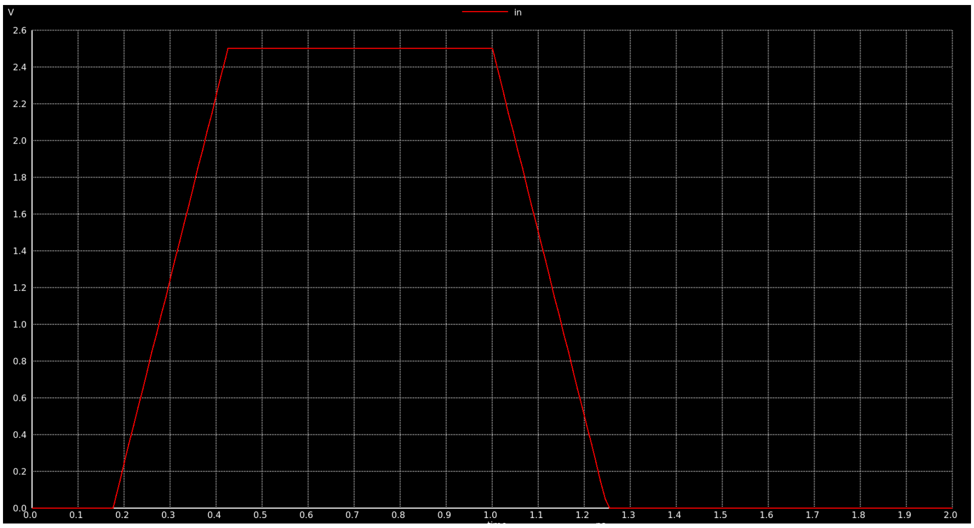

Figure 2: Transient response of the synthesized input signal V_{in}.

In accordance with the assignment instructions, the dynamic performance of the inverter is characterized by estimating the propagation delays (t_{pHL}, t_{pLH}) and the output transition times (t_{rise}, t_{fall}). To automate these measurements, the `.meas` command was utilized within the transient analysis.

The standard template provided in the instructions was adapted to capture specific transition events:

```
.meas tran <result_name>
+ trig v(node1) val=xx at=1
+ targ v(node2) val=xx at=1
```

For the purpose of this analysis, the `at=1` parameter was replaced with the `rise=1` and `fall=1` arguments to precisely target the first rising and falling edges of the signals, respectively.

For instance, the estimation of the high-to-low propagation delay (t_{pHL}) — defined from the 50% threshold of the rising input to the 50% threshold of the falling output — was implemented as follows:

```
.meas tran T_phl
+ trig v(in)  val=1.25 rise=1
+ targ v(out) val=1.25 fall=1
```

Similar measurement cards were constructed for t_{pLH} (using `fall=1` for input and `rise=1` for output) and for the output rise/fall times (measuring the interval between 10% and 90% of V_{DD} on the `out` node).

By executing the transient analysis and utilizing the `.meas` commands , the dynamic characteristics of the CMOS inverter were extracted. The simulation results for the propagation delays and the output transition times are summarized in Table 2.

Table 2: Extraction of Dynamic Timing Parameters from SPICE

Metric	**Result (s)**	**Targ (s)**	**Trig (s)**	**Final Value**
t_{pHL}	6.288×10^{-10}	8.538×10^{-10}	2.250×10^{-10}	$628.8\,ps$
t_{pLH}	7.455×10^{-10}	1.945×10^{-09}	1.200×10^{-09}	$745.5\,ps$
t_{rise}	5.624×10^{-10}	2.219×10^{-09}	1.656×10^{-10}	$562.4\,ps$
t_{fall}	2.871×10^{-10}	1.032×10^{-09}	7.449×10^{-10}	$287.1\,ps$

The dynamic response of the CMOS inverter is shown in Figure 3. We observe the output switching correctly between 2.5V (logic 1) and 0V (logic 0) over a 7ns simulation period.

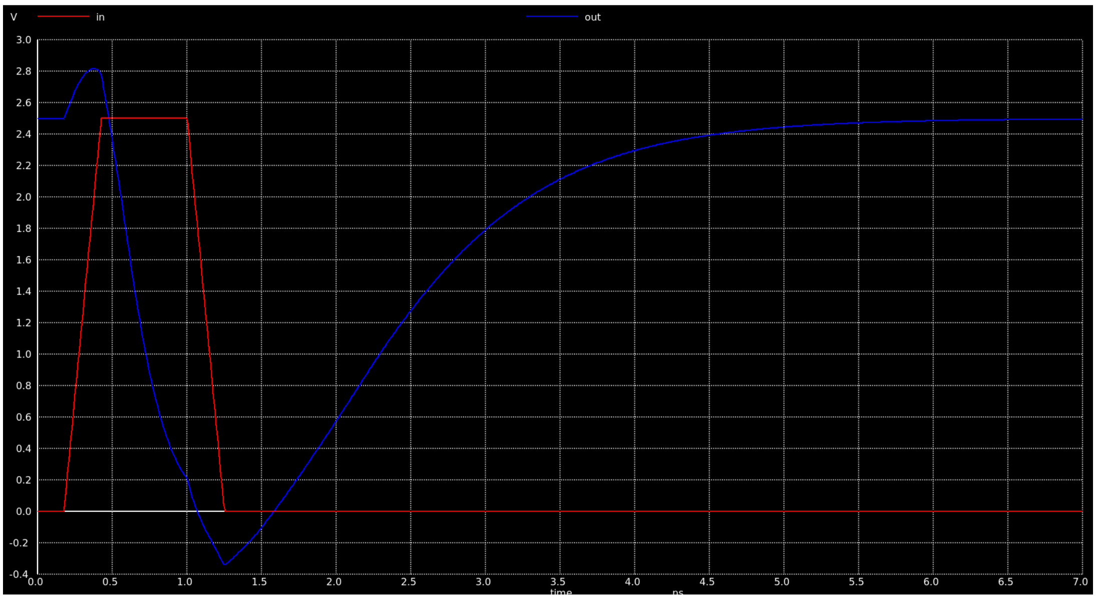

Figure 3: Transient response of the CMOS inverter illustrating input-to-output switching over a 7 ns period.